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Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
DT081A	T101	2026-08-18

Kursnamn	Datateknik AV, Signal- och bildbehandling
Provnamn	Tentamen
Ort	Sundsvall
Termin	
Ämne	

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2026-08-18. Examination in

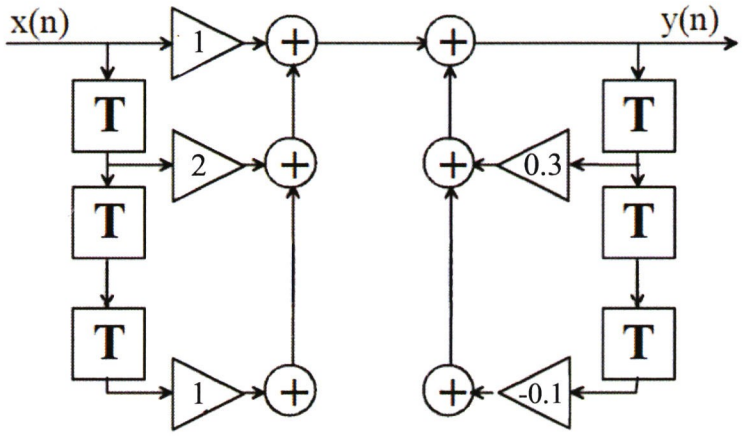
DT081A – Signal and Image Processing, 6 hp

Allowed aids	Dictionary, non-programmable calculator and drawing tools (e.g ruler), Physics Handbook, TeFyMa, and formula collection booklet for mathematics/physics courses at MIUN. A collection of formulas is also presented at the end of this exam.
Miscellaneous	Don't forget to: 1. Submit one, or several pages, per solved assignment and NOT several solved assignments per page. 2. Order the handed in pages with the solved assignments in ascending order. 3. Describe and clearly motivate all solutions, assumptions, programming constructs, etc.. 4. Attach all neatly drawn figures that you might want to refer to in a solution. **Failing to follow the above instructions will render point reduction.**

1. What do the following signal processing concepts mean? Be clear, concise, and relevant.

a) Sampling (1p)
b) Quantization (1p)
c) Impulse response (1p)
d) Frequency response (1p)
e) Windowing (1p)
f) Reconstruction (1p)

2. A time discrete system is defined as below.



a) Describe the system based on its impulse response behavior. (1p)
b) Present the system's difference equation. (2p)

3. Given a discrete-time system described by the difference equation

$$y(n) = x(n) - 0.5\,y(n-1) + 0.5\,y(n-2)$$

a) Identify the type of response that the system produces by the input

$$x(n) = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$ (1p)

b) Calculate the first 3 samples of the output signal $y(0), y(1), y(2)$ by the input

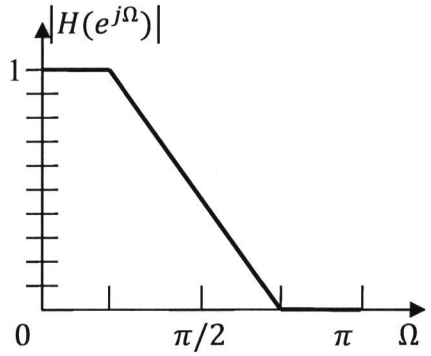
$$x[n] = \begin{cases} 0.1^n \cdot \cos[n], & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$ (2p)

c) Another system has the frequency function

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{1}{1-0.5e^{-j\Omega}-0.8e^{-2j\Omega}} = \frac{e^{2j\Omega}}{e^{2j\Omega}-0.5e^{j\Omega}-0.8}.$$

Sketch the amplitude magnification $|H(e^{j\Omega})|$ for frequency $0 \leq \Omega \leq 2\pi$. (3p)

4. A signal $s(t)$ have been sampled using $f_s = 100$ kHz. The amplitude spectrum of the sampled signal is shown in the figure below for a limited normalized frequency range.



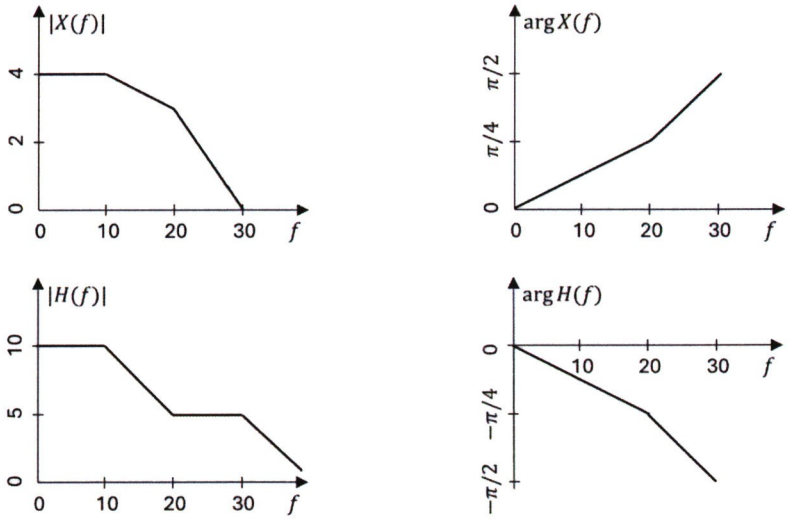
a) Sketch the amplitude spectrum for an extended normalized frequency range by extending axis to $-2\pi < \Omega < 2\pi$. (2p)

b) What is the lowest sampling frequency in Hz that the signal $s(t)$ could be sampled in while still being possible to be reconstructed exactly? Motivate. (3p)

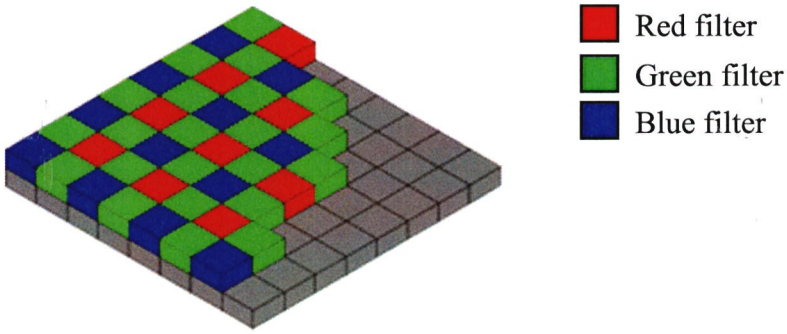
5. An LTI system $H(f)$ is defined by its amplification and phase-shift below. A signal $x(t)$ is fed to the system, giving the output $y(t)$.

a) Give the mathematical relation between amplitudes and phases of the input and the output. (2p)

b) Draw the amplitude spectrum and phase spectrum for the output $y(t)$. (4p)



6. Digital camera sensors typically capture only one color component (red, green, or blue) per pixel using a Bayer filter array like the one shown below. To reconstruct a full-color image, the missing color values at each pixel are interpolated from neighboring pixels.



a) Explain why the demosaicing (debayering) process necessarily involves a trade-off between color fidelity and spatial resolution. (2p)

b) Describe in general terms how a simple demosaicing algorithm reconstructs full RGB values from a Bayer-pattern image. *Hint: refer to how missing color values at each pixel are handled.* (2p)

7. The closeup image below has been filtered with 4 different filters.



Match the following kernels (1–5) with the corresponding filtered results (i–v). Justify your choices by explaining how the filter kernels affect the image.

Kernels

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$ **1)**

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$ **2)**

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$ **3)**

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$ **4)**

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$ **5)**

Filter results



i)



ii)



iii)



iv)



v)

(5p)

8. Contrast enhancement aims to improve the visual appearance of an image by modifying its intensity distribution. The figure below shows an original image (a) and an enhanced version (b). Plots (c–f) show different representations related to these two images. Describe what information each of the plots (c–f) conveys and explain, using equations where relevant, how they are involved in producing the enhanced image (b). (5p)

